State Estimation and Filtering for Robotic Perception

AA273

Course overview, Logistics,

Control Systems and Probability Review



Stanford University

MSL

Teaching Team

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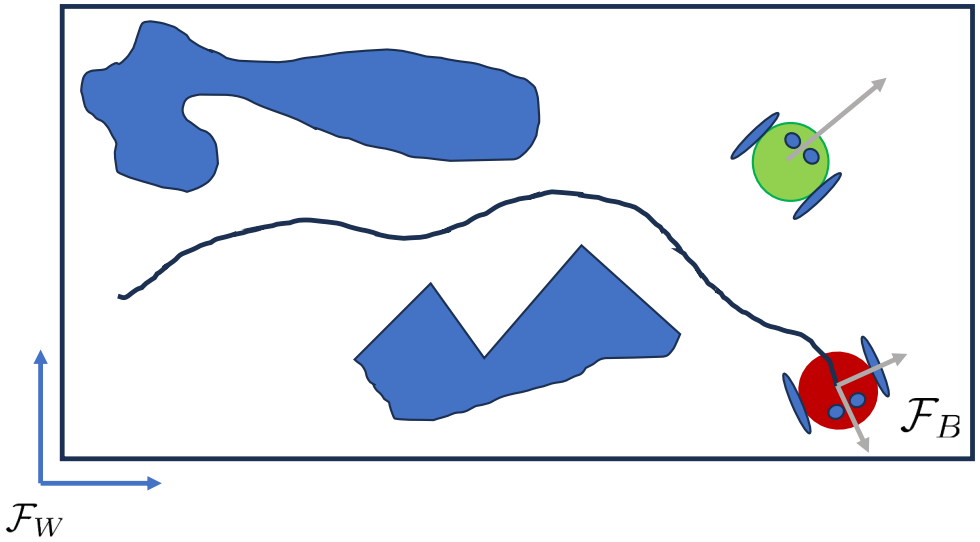


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State Estimation for Robotics: Overview



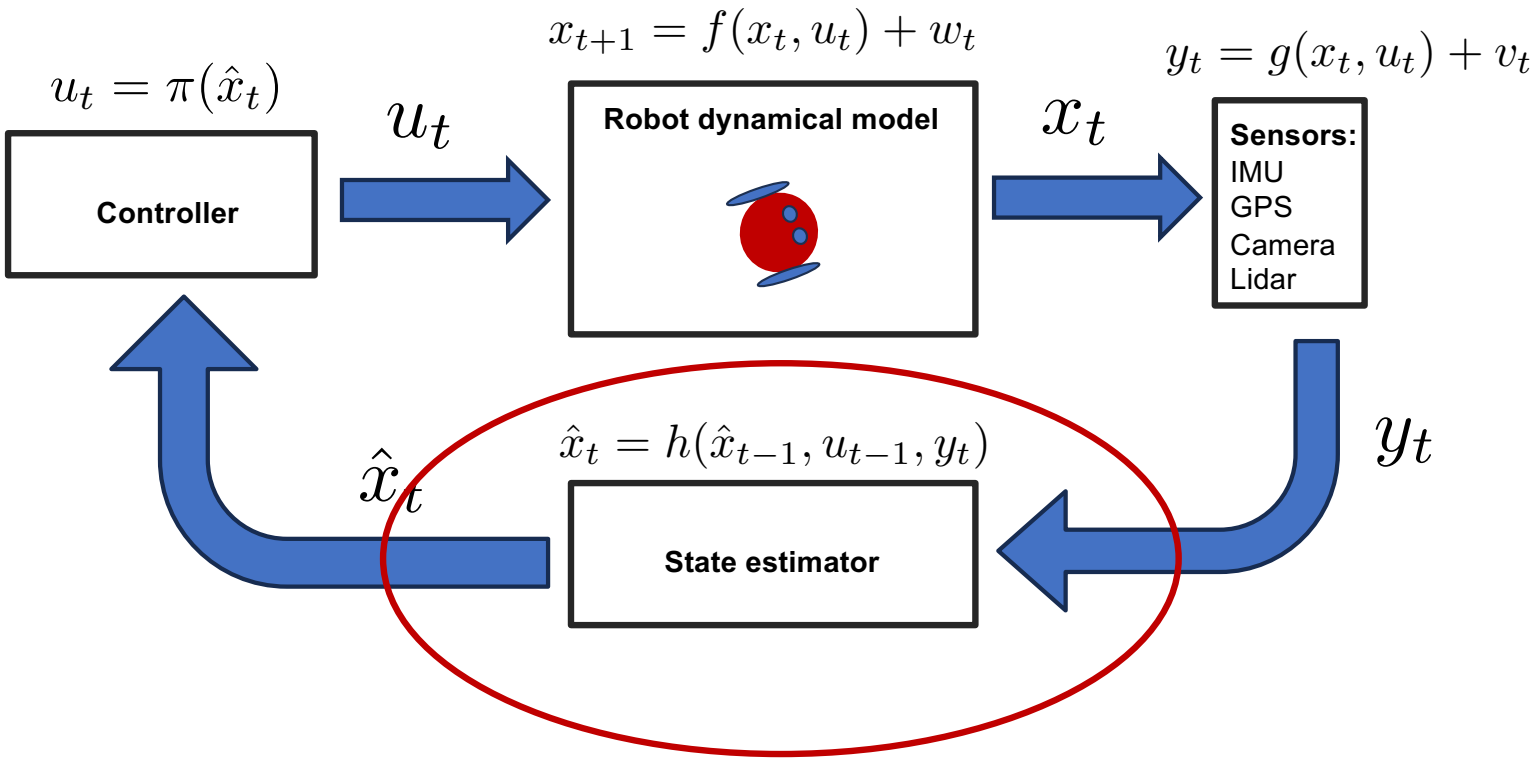
Where am I?
(Pose estimation/localization)

Where have I been?
(Trajectory estimation)

What are my surroundings?
(Mapping)

Where are the other agents going?
(Tracking)

Overview


$x_{t+1} = f(x_t, u_t) + w_t$
$y_t = g(x_t, u_t) + v_t$
$u_t = \pi(\hat{x}_t)$
u_t
Robot dynamical model
x_t
Controller
Sensors:
IMU
GPS
Camera
Lidar
y_t
$\hat{x}_t = h(\hat{x}_{t-1}, u_{t-1}, y_t)$
$\hat{x}_t$
State estimator

Overall Problem Statement (deterministic)

Given model:

Dynamics model: $f(\cdot, \cdot)$ Process noise distribution: $W_t \sim D(w_t)$

Sensor model: $g(\cdot, \cdot)$ Sensor noise distribution: $V_t \sim D(v_t)$

Given data:

Measurement history: $(y_1, \ldots, y_t)$ Control history: $(u_0, \ldots, u_{t-1})$

Find:

A state estimate: $\hat{x}_t$

Overall Problem Statement (probabilistic)

Given model:

Dynamics model: $f(\cdot, \cdot)$ Process noise distribution: $W_t \sim D(w_t)$

Sensor model: $g(\cdot, \cdot)$ Sensor noise distribution: $V_t \sim D(v_t)$

Given data:

Measurement history: $(y_1, \ldots, y_t)$ Control history: $(u_0, \ldots, u_{t-1})$

Find:

The probability distribution over the state: $b(x_t) := p(x_t \mid y_{1:t}, u_{0:t-1})$

$b(x_t)$ is called the belief, or the Bayesian posterior

Linear-Gaussian Systems and the Kalman Filter

For linear state space models with Gaussian white noise:

Dynamics model: $x_{t+1} = A_t x_t + B_t u_t + w_t$

Process noise: $w_t \sim GWN(0, Q_t)$

Sensor model: $y_t = C_t x_t + D_t u_t + v_t$

Sensor noise: $v_t \sim GWN(0, R_t)$

…and other technical conditions,

the Bayesian posterior is Gaussian: $\boxed{b(x_t) = \mathcal{N}(\mu_{t|t}, \Sigma_{t|t})}$

where the mean $\mu_{t|t}$ and covariance $\Sigma_{t|t}$ are found very efficiently with an algorithm called the **Kalman Filter**.

We'll see the details and precisely define notation in the coming lectures…

Course goals

To learn the *theoretical*, *algorithmic*, and *implementation* aspects of Kalman filtering, nonlinear filters, and Bayesian estimation techniques:

1. Mathematically rigorous framework for Bayesian state estimation for a robot or other dynamical system
2. Derivation and implementation for the Kalman filter, its performance guarantees, limitations, and underlying assumptions
3. Derivation and implementation of nonlinear, non-Gaussian filtering variants, EKF, UKF, PF
4. Optimization interpretation of state estimation
5. Application to robot mapping, pose estimation, SLAM, target tracking

Other robotics resources at Stanford

Manipulation Courses:

CS223A: Intro to robotics (Khatib)

CS225A: Experimental Robotics (Khatib)

CS327A: Advanced Manipulation (Khatib)

CS237B: Robot Autonomy II (Bohg, Sadigh, Pavone)

CS326: Advanced robotic manipulation (Bohg)

CS227: Robot perception (Song)

ME314: Robotic Dexterity (Kennedy)

Navigation Courses (including Localization, Mapping, SLAM):

CS237A: Robot Autonomy I (Schwager, Bohg)

CS237B: Robot Autonomy II (Bohg, Sadigh, Pavone)

AA203: Optimal and Learning-based Control (Pavone)

AA212: Advanced Feedback Control (Schwager)

AA275: Navigation for Autonomous Systems (Gao)

CS 224R: Deep RL (Finn)

ME 326: Collab Robotics (Kennedy)

CS 345: Building AI-Enabled Robots (Liu)

AA 276: Principles of Safety-Critical Autonomy (Bansal)

Stanford Robotics Center (SRC): https://src.stanford.edu/

Logistics

(the syllabus has all the info)

Structure

Lectures Students will be expected to attend lectures. Video recordings of the lectures will be available on the Canvas website.

Problem Sets Problem sets will be assigned regularly throughout the quarter. They will include a mix of traditional mathematical problems and implementation exercises in Matlab or Python. Of course, all solutions should contain your own original work. You can turn in your problem sets through the course gradescope page.

Midterm We will have a take-home midterm covering the material in the first half of the class. The midterm will be a 5 hour timed exam, which you can choose to take within a 24 hour window. More details will follow in class.

Project There will be a significant group project component to this class involving two steps. You will form groups of up to four students. First, groups will conduct a literature review in a relevant research area, and write a brief proposal for what they intend to do as a class project. Finally, groups will carry out the project, producing a written research report as well as an oral presentation for the class. Choosing a project that dovetails with your graduate research is highly encouraged.

Grading

All assignments should be submitted through the course gradescope page, and all grading will be done through gradescope. The final course grade will be assigned according to the following weighting:

Problem Sets		45%
Midterm		20%
Course project		35%
Project proposal	10%	
Final report & presentation	25%	

Websites

All course materials will be disseminated on the Canvas course website. A link to Canvas can be found at `https://canvas.stanford.edu/courses/`. We will also have an Ed page for class discussions, and a gradescope page for submitting and grading assignments.

Late Policy

- We assume you will turn in assignments on time.
- We will consider requests for late days that are submitted **at least 24 hours before the deadline** and with a legitimate justification: E.g., traveling for a conference, a serious illness, a major research or professional deadline or event.
- “I had a busy week and ran out of time,” is not a legitimate justification.
- Otherwise, we will deduct 10% from the assignment grade for each late day.

AI Assistance

As specified by the Stanford Honor Code on Generative AI Policy:

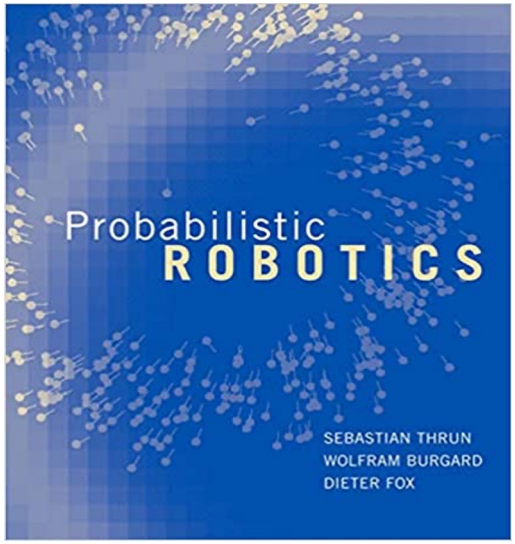
Absent a clear statement from a course instructor, use of or consultation with generative AI shall be treated analogously to assistance from another person. In particular, using generative AI tools to substantially complete an assignment or exam (e.g. by entering exam or assignment questions) is not permitted. Students should acknowledge the use of generative AI (other than incidental use) and default to disclosing such assistance when in doubt.

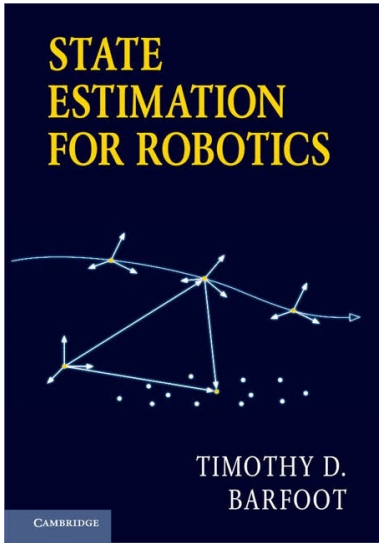
- Coding Assistants integrated into your IDE, Colab, Jupyter, Matlab environments **are allowed**. But you are responsible for understanding your code.
- Use of AI chat bots (e.g. ChatGPT, Gemini) is **not allowed** in any submitted assignments, except…
- You **may use** these AI tools to help generate illustrative figures or to format data tables for better readability. Carefully cite all such usage.
- If in doubt, ask a CA or me.
- Graders will be checking for unauthorized AI use.

OAE Letters

- If you have a letter, please email it to the CAs this week.
- Please alert the CAs before an assignment deadline if you plan to take OAE authorized late days to avoid misgrading or confusion.
- Please alert the CAs prior to an exam if you plan to use OAE accommodations so we can make appropriate arrangements.

Textbooks


Probabilistic
ROBOTICS
SEBASTIAN THRUN
WOLFRAM BURGARD
DIETER FOX


STATE
ESTIMATION
FOR ROBOTICS
TIMOTHY D.
BARFOOT
CAMBRIDGE

Schedule

Date	Topic	Reading	Assignments
1/6 T	Class Overview, Control Sys and Prob Review	ProRo 1	
1/8 Th	Rand Vars, Distributions, Bayes' Rule	ProRo 2, SER 2	PS 1 out
1/13 T	Rand Proc, Markov Proc, Graph Models	ProRo 2, SER 2	
1/15 Th	Bayesian Filter, multivariate Gaussian	ProRo 2, SER 2	PS 2 out, PS 1 due
1/20 T	Gaussian RVs, Moments, E[·], Cov(·)	ProRo 3.2, SER 2	
1/22 Th	Gaussian Estimation, Gaussian Processes	ProRo 3.2, SER 3.1	PS 3 out, PS 2 due
1/27 T	Linear Gaussian Systems	SER 3.3	
1/29 Th	Kalman Filter Derivation	ProRo 3.2, SER 3.1, 3.3	PS 4 out, PS 3 due
2/3 T	KF Properties, MMSE	ProRo 3.3, SER 3.3	
2/5 Th	Extended Kalman Filter (EKF, iEKF)	ProRo 3.4, SER 4.2	Midterm
2/10 T	Unscented Kalman Filter (UKF, iUKF)	ProRo 3.4, SER 4.2	
2/12 Th	Particle Filter (PF)	ProRo 4.3, SER 4.2	PS 5 out, PS 4 due
2/17 T	Non-Gaussian Est, MAP/ML Est	SER 4.3	Proposals due
2/19 Th	Rigid Body Poses, SLAM Front End/Back End	ProRo 6, SER 6.2, 6.3	PS 6 out, PS 5 due
3/24 T	Pose Graph Optimization Back End	ProRo 11, SER 8	
3/26 Th	Vision and Lidar Front Ends	ProRo 6, SER 6.4	
3/3 T	Camera Models, Rendering Equation		
3/5 Th	Special Topics: E.g. 3DGS, Diffusion Models		PS 6 due
3/10 Th	Project Presentations		
3/12 T	Project Presentations		Project reports due

Waitlisted students

- If you are on the waitlist and motivated to take the class, keep up with lectures and PSets for at least two weeks. There is a very good chance you will ultimately get into the class.
- Feel free to keep me updated on your position on the waitlist after lectures, and reiterate your interest in taking the class.
- I will find case-by-case solutions for those students still on the waitlist after two weeks, and still motivated to take the class.

Basic concepts in probability

- Key idea: quantities such as sensor measurements, states of a robot, and its environment are modeled as random variables (RVs)
- Discrete RV: the space of all the values that a random variable *X* can take on is *discrete*; characterized by probability mass function (pmf)

$$p(X = x) \quad (\text{or } p(x)), \qquad \sum_x p(X = x) = 1$$

Random variable — Specific value

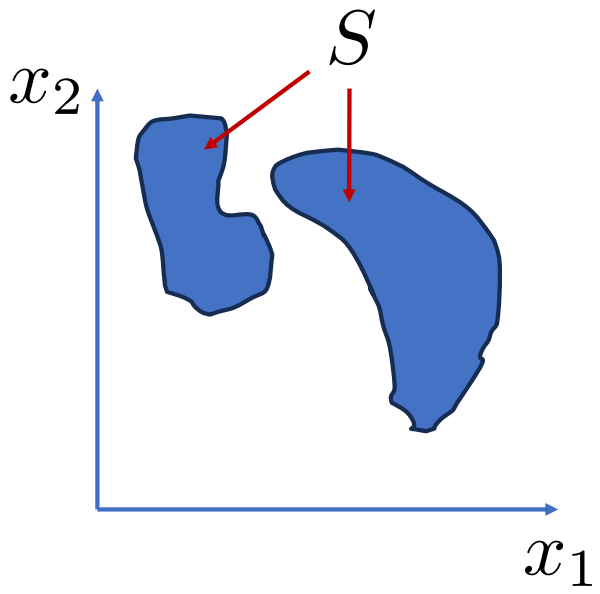
- Continuous RV: the space of all the values that a random variable *X* can take on is *continuous;* characterized by probability density function (pdf)

$$P(a \leq X \leq b) = \int_a^b p(x)\,dx, \quad P(X \in S) = \int_{x \in S} p(x)\,dx \qquad \int_{-\infty}^{\infty} p(x)\,dx = 1$$

Same for Random Vectors

$$X = [X_1 \quad X_2 \quad \cdots \quad X_n]^T \in \mathbb{R}^n$$

Pdf: $p(x) \geq 0 \quad \forall x \in \mathbb{R}^n$



$$P(X \in S) = \int_{x \in S} p(x)\, dx \qquad \int_{\mathbb{R}^n} p(x)\, dx = 1$$

Joint distribution, independence, and conditioning

- Joint distribution of two random variables *X* and *Y* is denoted as

$$p(x, y) := p(X = x \text{ and } Y = y)$$

- If *X* and *Y* are independent

$$p(x, y) = p(x)p(y)$$

- Suppose we know that $Y = y$ (with $p(y) > 0$); conditioned on this fact, the probability that the X's value is x is given by

$$p(x \mid y) := \frac{p(x, y)}{p(y)}$$

Conditional probability

Note: if *X* and *Y* are independent

$$p(x \mid y) := p(x)!$$